

Advanced Linear Algebra: Foundations to Frontiers

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2.3.6 Best rank-k approximation ¶

02.3.6 Best rank-k approximation

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We are now ready to answer the question "How do we find the best rank-k approximation for a picture (or, more generally, a matrix)?" posed in [Subsection 2.1.1](#).

Theorem 2.3.6.1. Given $A \in \mathbb{C}^{m \times n}$, let $A = U\Sigma V^H$ be its SVD. Assume the entries on the main diagonal of Σ are $\sigma_0, \dots, \sigma_{\min(m,n)-1}$ with $\sigma_0 \geq \dots \geq \sigma_{\min(m,n)-1} \geq 0$. Given k such that $0 \leq k \leq \min(m, n)$, partition

$$U = \left(U_L \mid U_R \right), V = \left(V_L \mid V_R \right), \text{ and } \Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & \Sigma_{BR} \end{array} \right),$$

where $U_L \in \mathbb{C}^{m \times k}$, $V_L \in \mathbb{C}^{n \times k}$, and $\Sigma_{TL} \in \mathbb{R}^{k \times k}$. Then

$$B = U_L \Sigma_{TL} V_L^H$$

is the matrix in $\mathbb{C}^{m \times n}$ closest to A in the following sense:

$$\|A - B\|_2 = \min_{\substack{C \in \mathbb{C}^{m \times n} \\ \text{rank}(C) \leq k}} \|A - C\|_2.$$

In other words, B is the matrix with rank at most k that is closest to A as measured by the 2-norm. Also, for this B ,

$$\|A - B\|_2 = \begin{cases} \sigma_k & \text{if } k < \min(m, n) \\ 0 & \text{otherwise.} \end{cases}$$

The proof of this theorem builds on the following insight:

Homework 2.3.6.1. Given $A \in \mathbb{C}^{m \times n}$, let $A = U\Sigma V^H$ be its SVD. Show that

$$Av_j = \sigma_j u_j \text{ for } 0 \leq j < \min(m, n),$$

where u_j and v_j equal the columns of U and V indexed by j , and σ_j equals the diagonal element of Σ indexed with j .

▼ Solution

W.l.o.g. assume $n \leq m$. Rewrite $A = U\Sigma V^H$ as $AV = U\Sigma$. Then

$$\begin{aligned} AV = U\Sigma &= \text{< partition >} \\ A \left(v_0 \mid \cdots \mid v_{n-1} \right) &= \left(u_0 \mid \cdots \mid u_{n-1} \mid u_n \mid \cdots \mid u_{m-1} \right) \begin{pmatrix} \sigma_0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_{n-1} \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & & 0 \end{pmatrix} \\ &= \text{< multiply out >} \\ \left(Av_0 \mid \cdots \mid Av_{n-1} \right) &= \left(\sigma_0 u_0 \mid \cdots \mid \sigma_{n-1} u_{n-1} \right). \end{aligned}$$

Hence $Av_j = \sigma_j u_j$ for $0 \leq j < n$.

Proof of Theorem 2.3.6.1.

First, if B is as defined, then $\|A - B\|_2 = \sigma_k$:

$$\begin{aligned}
& \|A - B\|_2 \\
&= \quad < \text{multiplication with unitary matrices preserves 2-norm} > \\
& \|U^H(A - B)V\|_2 \\
&= \quad < \text{distribute} > \\
& \|U^H A V - U^H B V\|_2 \\
&= \quad < \text{use SVD of } A \text{ and partition} > \\
& \left\| \Sigma - \begin{pmatrix} U_L & U_R \end{pmatrix}^H B \begin{pmatrix} V_L & V_R \end{pmatrix} \right\|_2 \\
&= \quad < \text{how } B \text{ was chosen} > \\
& \left\| \begin{pmatrix} \Sigma_{TL} & 0 \\ 0 & \Sigma_{BR} \end{pmatrix} - \begin{pmatrix} \Sigma_{TL} & 0 \\ 0 & 0 \end{pmatrix} \right\|_2 \\
&= \quad < \text{partitioned subtraction} > \\
& \left\| \begin{pmatrix} 0 & 0 \\ 0 & \Sigma_{BR} \end{pmatrix} \right\|_2 \\
&= \quad < > \\
& \|\Sigma_{BR}\|_2 \\
&= \quad < \Sigma_{TL} \text{ is } k \times k > \\
& \sigma_k
\end{aligned}$$

(Obviously, this needs to be tidied up for the case where $k > \text{rank}(A)$.)

Next, assume that C has rank $r \leq k$ and $\|A - C\|_2 < \|A - B\|_2$. We will show that this leads to a contradiction.

- The null space of C has dimension at least $n - k$ since $\dim(\mathcal{N}(C)) = n - r$.
- If $x \in \mathcal{N}(C)$ then

$$\|Ax\|_2 = \|(A - C)x\|_2 \leq \|A - C\|_2 \|x\|_2 < \sigma_k \|x\|_2.$$

- Partition $U = \begin{pmatrix} u_0 & \cdots & u_{m-1} \end{pmatrix}$ and $V = \begin{pmatrix} v_0 & \cdots & v_{n-1} \end{pmatrix}$. Then $\|Av_j\|_2 = \|\sigma_j u_j\|_2 = \sigma_j \geq \sigma_k$ for $j = 0, \dots, k$.
- Now, let y be any linear combination of $v_0, \dots, v_k$:
 $y = \alpha_0 v_0 + \cdots + \alpha_k v_k$. Notice that

$$\|y\|_2^2 = \|\alpha_0 v_0 + \cdots + \alpha_k v_k\|_2^2 = |\alpha_0|^2 + \cdots + |\alpha_k|^2$$

since the vectors v_j are orthonormal. Then

$$\begin{aligned}
& \|Ay\|_2^2 \\
&= \langle y = \alpha_0 v_0 + \dots + \alpha_k v_k \rangle \\
& \|A(\alpha_0 v_0 + \dots + \alpha_k v_k)\|_2^2 \\
&= \langle \text{distributivity} \rangle \\
& \|\alpha_0 A v_0 + \dots + \alpha_k A v_k\|_2^2 \\
&= \langle A v_j = \sigma_j u_j \rangle \\
& \|\alpha_0 \sigma_0 u_0 + \dots + \alpha_k \sigma_k u_k\|_2^2 \\
&= \langle \text{this works because the } u_j \text{ are orthonormal} \rangle \\
& \|\alpha_0 \sigma_0 u_0\|_2^2 + \dots + \|\alpha_k \sigma_k u_k\|_2^2 \\
&= \langle \text{norms are homogeneous and } \|u_j\|_2 = 1 \rangle \\
& |\alpha_0|^2 \sigma_0^2 + \dots + |\alpha_k|^2 \sigma_k^2 \\
&\geq \langle \sigma_0 \geq \sigma_1 \geq \dots \geq \sigma_k \geq 0 \rangle \\
& (|\alpha_0|^2 + \dots + |\alpha_k|^2) \sigma_k^2 \\
&= \langle \|y\|_2^2 = |\alpha_0|^2 + \dots + |\alpha_k|^2 \rangle \\
& \sigma_k^2 \|y\|_2^2.
\end{aligned}$$

so that $\|Ay\|_2 \geq \sigma_k \|y\|_2$. In other words, vectors in the subspace of all linear combinations of $\{v_0, \dots, v_k\}$ satisfy $\|Ax\|_2 \geq \sigma_k \|x\|_2$. The dimension of this subspace is $k+1$ (since $\{v_0, \dots, v_k\}$ form an orthonormal basis).

- Both these subspaces are subspaces of $\mathbb{C}^n$. One has dimension $k+1$ and the other $n-k$. This means that if you take a basis for one (which consists of $n-k$ linearly independent vectors) and add it to a basis for the other (which has $k+1$ linearly independent vectors), you end up with $n+1$ vectors. Since these cannot all be linearly independent in $\mathbb{C}^n$, there must be at least one nonzero vector z that satisfies both $\|Az\|_2 < \sigma_k \|z\|_2$ and $\|Az\|_2 \geq \sigma_k \|z\|_2$, which is a contradiction.

Theorem 2.3.6.1 tells us how to pick the best approximation to a given matrix of a given desired rank. In Section **Subsection 2.1.1** we discussed how a low rank matrix can be used to compress data. The SVD thus gives the best such rank- k approximation. Let us revisit this.

Let $A \in \mathbb{R}^{m \times n}$ be a matrix that, for example, stores a picture. In this case, the i, j entry in A is, for example, a number that represents the grayscale value of pixel (i, j) .

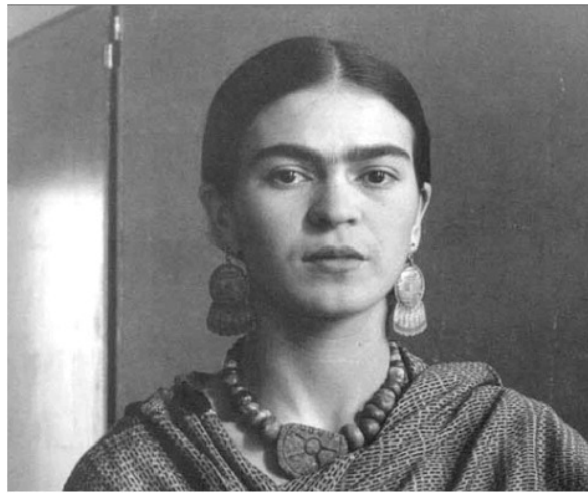
Homework 2.3.6.2. In Assignments/Week02/matlab execute

```

IMG = imread( 'Frida.jpg' );
A = double( IMG( :, :, 1 ) );
imshow( uint8( A ) )
size( A )

```

to generate the picture of Mexican artist Frida Kahlo



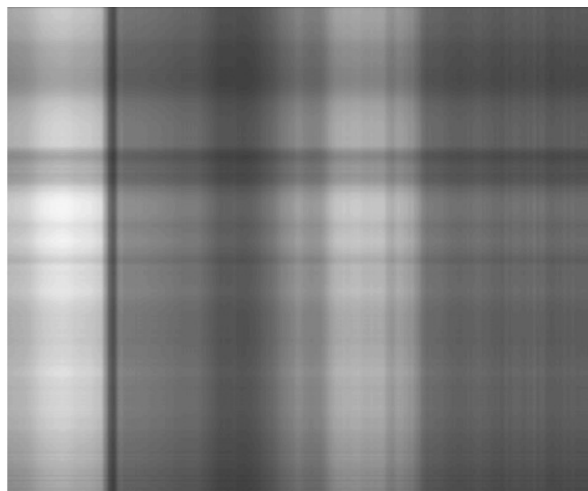
Although the picture is black and white, it was read as if it is a color image, which means a $m \times n \times 3$ array of pixel information is stored. Setting $A = \text{IMG}(:, :, 1)$ extracts a single matrix of pixel information. (If you start with a color picture, you will want to approximate $\text{IMG}(:, :, 1)$, $\text{IMG}(:, :, 2)$, and $\text{IMG}(:, :, 3)$ separately.)

Next, compute the SVD of matrix A

```
[ U, Sigma, V ] = svd( A );
```

and approximate the picture with a rank- k update, starting with $k = 1$:

```
k = 1
B = uint8( U( :, 1:k ) * Sigma( 1:k, 1:k ) * V( :, 1:k )' );
imshow( B );
```



Repeat this with increasing k .

```
r = min( size( A ) )
for k=1:r
    imshow( uint8( U( :, 1:k ) * Sigma( 1:k, 1:k ) * V( :, 1:k )' ) );
    input( strcat( num2str( k ), "    press return" ) );
end
```

To determine a reasonable value for k , it helps to graph the singular values:

```
figure
r = min( size( A ) );
```

```
plot( [ 1:r ], diag( Sigma ), 'x' );
```

Since the singular values span a broad range, we may want to plot them with a log-log plot

```
loglog( [ 1:r ], diag( Sigma ), 'x' );
```

For this particular matrix (picture), there is no dramatic drop in the singular values that makes it obvious what k is a natural choice.

▼ Solution

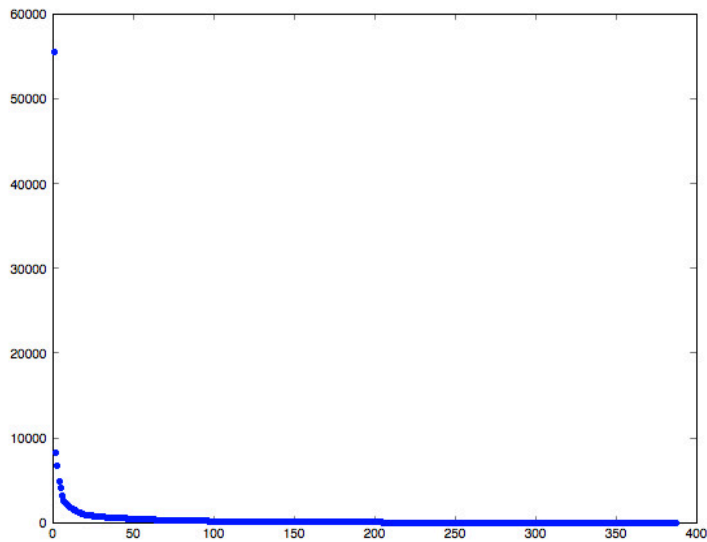
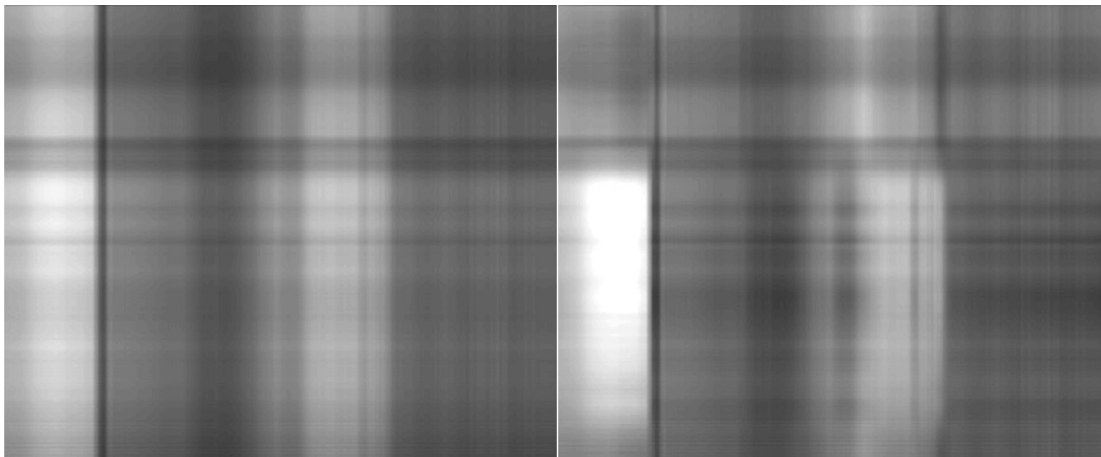


Figure 2.3.6.2. Distribution of singular values for the picture.

$k = 1$

$k = 2$



$k = 5$

$k = 10$



$k = 25$

Original picture

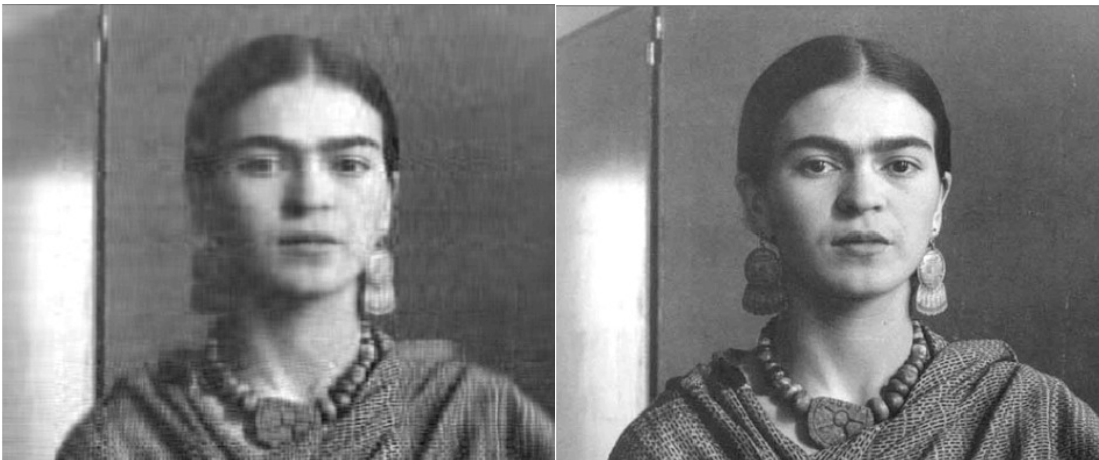


Figure 2.3.6.3. Multiple pictures as generated by the code.